Chapter 11: Work and Energy

In daily life we call any tiring activity "work", whether it is reading, thinking or standing with a load. In physics the word has a narrow, exact meaning that links force and displacement. Energy is the capacity to do work, and power tells us how quickly that work is done. These notes cover the five topics of the chapter with solved examples for practice.

11.1 Work

Work is said to be done on an object only when a force acts on it and the object is displaced. Both conditions are necessary: a force must act, and the object must move with some part of its displacement along the force.

When a constant force F moves an object through a displacement s in the direction of the force, the work done is W = F × s. Work is a scalar quantity; it has magnitude but no direction.

The SI unit of work is the joule (J). One joule is the work done when a force of 1 newton moves an object through 1 metre in the direction of the force, so 1 J = 1 N m. A larger unit is the kilojoule: 1 kJ = 1000 J.

If the force acts at an angle θ to the displacement, only the part of the force along the displacement does work, and W = F s cos θ.

- Positive work (θ = 0°): force and displacement are in the same direction, as when gravity pulls a falling mango downward.
- Negative work (θ = 180°): force opposes the displacement, as when friction acts on a sliding box, or when gravity acts on a ball thrown upward.
- Zero work: there is no displacement, as when a man pushes a rigid wall and it does not move.
- Zero work (θ = 90°): force is perpendicular to the displacement, as when a porter carries a load on his head along a level road; gravity does no work on the load.

Worked example: A labourer pushes a handcart with a steady force of 150 N along a straight road for 20 m, in the direction of motion. Work done = F × s = 150 × 20 = 3000 J = 3 kJ.

Worked example: A box slides 5 m on a floor while a friction force of 10 N acts opposite to its motion. Work done by friction = −10 × 5 = −50 J.

11.2 Kinetic Energy

Energy is the capacity of an object to do work. Its SI unit is also the joule. When an object does work it loses energy, and the object on which the work is done gains the same amount. Energy appears in many forms: mechanical, heat, light, sound, chemical, electrical and nuclear. Mechanical energy is the sum of kinetic energy and potential energy.

Kinetic energy is the energy an object possesses because of its motion. A moving bullet, flowing river water that turns a water wheel, wind that turns the blades of a windmill and a swinging hammer that drives a nail all have kinetic energy.

Formula: an object of mass m moving with speed v has kinetic energy KE = ½ m v^2. The SI unit is the joule.

Derivation: let a constant force F act on an object of mass m and change its velocity from u to v over a displacement s. From the equation of motion, $v^2 - u^2 = 2as$, so $s = (v^2 - u^2)/2a$. The work done is $W = F \times s = ma \times (v^2 - u^2)/2a = ½ m (v^2 - u^2)$. If the object starts from rest (u = 0), W = ½ m v^2, and this work is stored as its kinetic energy.

Work-energy theorem: the work done by the net force on an object equals the change in its kinetic energy, W = ½ m v^2 − ½ m u^2.

- Kinetic energy is directly proportional to mass and to the square of speed.
- If the speed is doubled, kinetic energy becomes four times; if the speed is halved, it becomes one-fourth.
- Kinetic energy can never be negative, because m is positive and v^2 cannot be negative.
- In terms of momentum p = mv, kinetic energy can also be written as KE = $p^2/2m$.
- Of two bodies with equal momentum, the lighter body has the greater kinetic energy, because KE = $p^2/2m$.

Worked example: A cricket ball of mass 0.16 kg is hit at 25 m/s. KE = ½ × 0.16 × 25^2 = 0.08 × 625 = 50 J.

Worked example: How much work is needed to speed up a car of mass 800 kg from 10 m/s to 15 m/s? W = ½ × 800 × (15^2 − 10^2) = 400 × (225 − 100) = 400 × 125 = 50000 J = 50 kJ.

11.3 Potential Energy

Potential energy is the energy an object possesses because of its position or its configuration (shape). Water stored behind a dam, a stretched bow, a stretched rubber band, the wound spring of a toy car and a hammer held up high all store potential energy.

Gravitational potential energy: to raise an object of mass m through a height h, a force equal to its weight mg must act through the height h. The work done, mg × h, is stored in the object. So gravitational potential energy is PE = mgh, where g = 9.8 m/s² (taken as 10 m/s² in many numericals). The height is measured from a chosen zero level, usually the ground.

- Gravitational potential energy depends only on the vertical height gained, not on the path followed. Lifting a box by 2 m up a staircase or up a smooth ramp stores the same energy.
- Potential energy is directly proportional to height: doubling the height doubles the potential energy.
- Elastic potential energy is stored when the shape of an object is changed, as in a compressed spring, a bent bow or a stretched catapult (gulel).

Worked example: A bucket of water of mass 5 kg is lifted to a roof 8 m high. Taking g = 10 m/s², PE = mgh = 5 × 10 × 8 = 400 J.

11.4 Conservation of Energy

Law of conservation of energy: energy can neither be created nor destroyed; it can only be changed from one form to another. The total energy before and after any transformation stays the same.

For an object falling freely (air resistance neglected), potential energy turns into kinetic energy, but the sum KE + PE, the total mechanical energy, stays constant at every point. Take a 2 kg stone dropped from a height of 20 m, with g = 10 m/s².

- At the top (h = 20 m): PE = 2 × 10 × 20 = 400 J, KE = 0, total = 400 J.
- After falling 10 m (h = 10 m): PE = 200 J; v² = 2 × 10 × 10 = 200, so KE = ½ × 2 × 200 = 200 J; total = 400 J.
- Just before touching the ground (h = 0): PE = 0, v = 20 m/s, KE = 400 J, total = 400 J.

On striking the ground, the kinetic energy changes into heat, sound and work done in denting the ground; the energy is not lost.

Energy Transformations Around Us

- Simple pendulum: at the extreme positions the bob is momentarily at rest and its energy is all potential; at the mean position its kinetic energy is maximum. It finally stops because air resistance and friction turn its mechanical energy into heat.
- Hydroelectric station, such as the one at Rana Pratap Sagar dam on the Chambal: potential energy of stored water changes into kinetic energy and then into electrical energy.
- Electric cell: chemical energy to electrical energy. Solar cell: light energy to electrical energy.
- Electric bulb: electrical energy to light and heat. Electric motor: electrical to mechanical energy.

11.5 Power

Power is the rate of doing work, or the rate of transfer of energy: P = W / t. Power is a scalar quantity. Its SI unit is the watt (W): 1 W = 1 J/s, the power of an agent that does 1 joule of work in 1 second. Larger units are 1 kW = 1000 W and 1 MW = 1000 kW. An older practical unit is the horsepower: 1 hp = 746 W.

When the rate of doing work changes with time, we use average power = total work done ÷ total time taken.

Worked example: A student of mass 40 kg runs up a staircase 6 m high in 12 s. Work done = mgh = 40 × 10 × 6 = 2400 J, so power = 2400 ÷ 12 = 200 W.

Commercial unit of energy: electricity bills charge for energy in kilowatt hours (kWh), called "units". 1 kWh is the energy used in one hour at the rate of 1 kW. 1 kWh = 1000 W × 3600 s = 3600000 J = 3.6 × 10^6 J.

Worked example: A 400 W desert cooler runs 10 hours a day for 30 days. Energy = 0.4 kW × 10 h × 30 = 120 kWh = 120 units. At Rs 6 per unit, the bill is Rs 720.

Key Facts at a Glance

- Work and energy are scalars measured in joules; power is a scalar measured in watts.
- W = F s cos θ; KE = ½ m v^2; PE = mgh; P = W / t.
- 1 kWh = 3.6 × 10^6 J and 1 hp = 746 W.
- The kilowatt hour is a unit of energy, not of power.